CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M.Tech. (EC) Examination

December- 2012

EC – 701 Advanced High Speed Networks

Date: 21.12.2012, Friday **Time: 10.00 a.m to 1.00 p.m** **Maximum Marks: 70**

Instructions: **(1)** Attempt all questions.
(2) Answer to the two sections must be written in separate answer books.
(3) Figures to the right indicate full marks
(4) Make suitable assumptions and draw neat figures whenever required.

SECTION –I

Q-1(a)	Give three differences between X.25 and Frame Relay.	**03**
(b)	What do you mean by internetworking? Discuss the function of router.	**04**
Q-2(a)	With neat sketch discuss the basic architecture of ATM for an interface between user and network.	**07**
(b)	Give the characteristics of some high speed LANs. Explain Gigabit Ethernet in detail.	**07**
	OR	
Q-2 (a)	Discuss Fiber Channel Protocol Architecture in Brief.	**05**
(b)	Give the difference between datagram circuit switching and virtual circuit approach for packet switching techniques.	**05**
(c)	Suppose that an 11- Mbps 802.11b LAN is transmitting 64-byte frames back-to-back over a radio channel with bit error rate of 10^{-7}. How many frames per second will be damaged on average?	**04**
Q-3 (a)	Explain the Data Encryption Standard (DES) in detail.	**07**
(b)	What is the need of protocol architecture? Explain the architecture of frame relay.	**07**
	OR	
Q-3(a)	Describe transposition cipher for computer network security	**05**
(b)	Write a short note on ALL services	**05**
(c)	Explain the operation of TCP/IP reference model with figure.	**04**

SECTION-II

Q-4 (a)	Describe count – to – infinity problem. In detail, describe two solutions to overcome it.	04
(b)	Give at least three differences between Distance–Vector routing and Link State routing.	03
Q-5(a)	Explain the need of TCP flow control. With neat sketch explain example of TCP credit allocation mechanism.	07
(b)	With necessary equations describe RTT Variance Estimation	07
	OR	
Q-5 (a)	With neat illustration explain RSVP Operation.	07
(b)	Describe MPLS Packet forwarding using diagrams.	07
Q-6(a)	Draw neat diagram of Integrated Services Architecture (ISA) implementation in Router and explain ISA components.	07
(b)	Explain Real – time Transport Protocol (RTP) Protocol Architecture.	07
	OR	
Q-6(a)	Explain Karn's Algorithm for TCP Traffic Control.	05
(b)	Describe Real – time Transport Protocol (RTP) Header.	05
(c)	Express Guaranteed, Controlled Load and Best effort Services implemented in Integrated Service Architecture.	04
